H

NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 2

MATHEMATICS 9758/02

Paper 2 **11 September 2018**

3 hours

Additional Materials: Answer Paper
List of Formulae (MF26)
Cover Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, registration number, subject tutorial group, on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

Section A: Pure Mathematics [40 marks]

1 **(a)** In a city, a traveller may use a mobile phone application to choose a variety of modes of transport to travel from the international airport A, to a specific hotel Z: taking a High Speed Rail, taking a Subway train or walking.

Along the way, he will need to make transfer via various interchange stations or subway stations B, C, D, E, F or G and walk within stations to travel in another mode of transport.

When the traveller uses the travelling application, he is given 3 choices to travel from the international airport to the hotel:

<table>
<tr><th></th><th colspan="4">International Airport ⟶ Hotel</th></tr>
<tr><td>Choice I</td><td rowspan="2">Take the High Speed Rail at A for 60km to Interchange Station B</td><td>Walk 0.3 km to Subway Station within B</td><td>Take the Subway train within B to C that travels 10 km.</td><td>Walks 0.5 km to Z.</td></tr>
<tr><td>Choice II</td><td>Within B, cross over to another platform at a negligible distance away to take the Subway train and travel 5 km to D.</td><td>Walks 1.2 km to Z.</td><td></td></tr>
<tr><td>Choice III</td><td>Takes the Subway train at A for 20 km to Interchange Station E.</td><td>Within E, walks 0.2 km to F.</td><td>Take the Subway train at F to G for 40 km</td><td>Walks 0.2 km to Z.</td></tr>
</table>

Given that the travelling time taken for Choices I, II and III are 51.7 minutes, 56.3 minutes and 69.6 minutes and assuming that the waiting time for the high speed rail or subway is negligible, find the average speed of the high speed rail, subway train and the walking pace of the traveller. Leave your answers in km/h. [4]

(b) **(i)** Solve the inequality $\dfrac{2x-1}{2x^2-1} \le 1$ exactly. [3]

(ii) Hence, solve the inequality $\dfrac{2(\cos x)-1}{\cos 2x} \le 1$ exactly for $0 \le x \le \pi$. [3]

2 The planes p_1 and p_2, have equations $2x+3y+6z=0$ and $\mathbf{r} \cdot \begin{pmatrix}1\\2\\2\end{pmatrix}=6$ respectively.

(i) Find a vector equation of the line of intersection, *l*, between p_1 and p_2. [2]

The line *m* passes through the points $A(2,1,1)$ and $B(5,4,2)$.

(ii) Verify that *A* lies on p_2. [1]

(iii) Find the coordinates of the points on *m* that are equidistant from planes p_1 and p_2. [5]

3 **(a)** The sum, S_n, of the first *n* terms of a sequence is given by

$$S_n=\pi\left(n-\frac{1}{2}\right)^2-\frac{\pi}{4}+n\pi^2.$$

Show that the sequence follows an arithmetic progression and state the value of the common difference. [5]

(b) **(i)** A kangaroo spotted her runaway joey when she was 20 metres away from her and jumped straight towards her joey. The kangaroo's 1st jump was 2 metres and each successive jump was 1.1 times that of her previous jump. At the same time, the joey jumped away from her mother along the same imaginary straight line, such that its first jump was 1 metre and each successive jump was 0.1 metres more than its previous jump.

Assuming that the time taken for the kangaroo and her joey for each jump is the same, set up an inequality for the least number of jumps, *k*, the kangaroo had to take in order to catch up with her joey. Hence, find the value of *k*. [4]

(ii) Find the distance between the kangaroo and her joey just after the *k*th jump. [1]

4 **(a)** It is given that $z^* = \dfrac{(-1-\mathrm{i})^3}{1-\mathrm{i}\sqrt{3}}$, where z^* is the conjugate of a complex number z.

(i) Find the exact values of the modulus and argument of $\dfrac{1}{z}$. [5]

(ii) Hence determine the exact values of a and b (where $-\pi < b \le \pi$) in the equation

$$\mathrm{e}^{2a+\mathrm{i}b} = \frac{1}{z^4}.$$ [3]

(b) The complex variables u and v satisfy the equations

$$\mathrm{i}u - v = 3 \quad \text{and} \quad u^* + (1-\mathrm{i})v = 7 + 4\mathrm{i}.$$

Find the values of u and v, giving your answers in the form $x + \mathrm{i}y$. [4]

Section B: Probability and Statistics [60 marks]

5 In this question you should state clearly the values of the parameters of any normal distribution you use.

The masses in grams of dragon fruits and mangoes are independent random variables with the distributions $\mathrm{N}(350, 196)$ and $\mathrm{N}(250, 98)$ respectively. Dragon fruits and mangoes are priced at \$2.40 per kilogram and \$12 per kilogram respectively.

Find the probability that the total cost of 5 randomly chosen dragon fruits and 3 randomly chosen mangoes is less than \$13.50. [4]

6 A school's concert band comprises 24 woodwind players, n brass players and 10 percussion players. $\dfrac{1}{3}$ of all woodwind players, $\dfrac{2}{5}$ of all brass players and $\dfrac{4}{5}$ of all percussion players are Senior High students, while the rest are Junior High students. No student in the band plays more than one type of instrument.

One student from the concert band is selected at random. Find, in terms of n, the probability that the student is neither a percussion player nor a Senior High student. [3]

Suppose instead that two students from the concert band are randomly selected. It is given that the probability that one of them is a Senior High student and the other is a Junior High student is $\dfrac{1}{2}$. Show that

$$n^2 + pn + q = 0$$

for some integer constants p and q to be determined, and hence find the value of n. [3]

7 Ten circular stickers for temperature taking purposes, each of them indistinguishable apart from their colours, are placed in an opaque box.

(a) It is given that four of the stickers are purple, two are blue and the remaining stickers are pink, orange, yellow and green. Suppose four stickers are given to 4 people, such that each person receives exactly one sticker. Find the number of ways this can be done if

(i) all four stickers are of different colours, [1]

(ii) there are no restrictions on the colours of the stickers. [3]

(b) The 10 stickers labelled with distinct alphabets "A" to "J" are to be packed into zip-lock bags, which may come in different sizes. Bags of the same size are considered to be indistinguishable.

(i) Suppose five zip-lock bags of different sizes are used to contain 2 stickers each. How many ways can this be done? [1]

(ii) Suppose instead that a large-sized bag is used to contain 4 stickers, two medium-sized bags are used to contain 2 stickers each and a small-sized bag is used to contain the remaining 2 stickers. How many ways can this be done? [2]

8 Mary and Jerry play a game using two six-sided dice. One of the dice is fair with '1' to '6' on each face. The other die is weighted such that the score, denoted by Y, has a probability distribution given as follows.

$$\mathrm{P}(Y=y)=\begin{cases}\dfrac{1}{6} & \text{for } y=1,\\[2mm] \dfrac{1}{6}(y-1) & \text{for } y=2,3,\\[2mm] \dfrac{1}{36}(y-1) & \text{for } y=4,5,6.\end{cases}$$

Mary throws the dice. Jerry pays Mary \$3 if the total score from the two dice is at least "9" and Mary pays Jerry \$4 if the total score from the two dice is at most "4". Otherwise, there is no monetary transaction between both parties. Let \$$X$ be the amount of money Mary gains after one game.

Show that $\mathrm{P}(X=3)=\dfrac{25}{108}$. Tabulate the probability distribution of X. [4]

The game is played n times, where n is large. Find the least number of games that Mary needs to play so that there is a probability of more than 0.99 for her average winnings in the n games to differ from her expected winnings by at most \$1. State clearly the parameters used in your calculations. [5]

9 Factory Nayma produces balls used by the World Ball Association with diameters that follow a normal distribution. A random sample of 30 balls is chosen and their diameters in centimetres, x, are summarised by

$$\sum(x-22) = -27 \text{ and } \sum(x-22)^2 = 321.26.$$

(i) Calculate unbiased estimates of the population mean and population variance. [2]

(ii) Test, at the 10% significance level, whether the population mean diameter of a ball produced by Factory Nayma is 22 cm. [5]

(iii) Another random sample of 30 balls is selected, with mean diameter $\overline{x}$. Use an algebraic method to calculate the set of values of $\overline{x}$ for which there is insufficient evidence to conclude that the population mean diameter of all balls is greater than 22 cm at the 10% level of significance. (Answers obtained by trial and improvement from a calculator will obtain no marks.) [3]

10 5.2% of all insurance agents from a large insurance company, Prodential, have an advanced diploma in insurance (ADI). A random sample of 30 agents from Prodential is obtained.

(i) State, in context, two assumptions for the number of insurance agents with ADI in this sample to be well-modelled by a binomial distribution. [2]

Assume now that these assumptions do in fact hold.

(ii) Find the probability that at least three of the insurance agents in this sample have an ADI each. [2]

100p% of all insurance agents from another large insurance company, Avila, have an advanced diploma in insurance (ADI), where $p < 0.5$. A sample of 10 agents from Avila is obtained. It is given that the number of insurance agents with ADI in this sample can be modelled by a binomial distribution.

(iii) It is given that the probability that 5 of the agents in this sample have an ADI each is 0.12294, correct to 5 decimal places. Show that p satisfies an equation of the form $p(1-p) = k$ for some real constant k to be determined, and hence find the value of p correct to 2 decimal places. [3]

(iv) Suppose instead that $p = 0.24$ and forty samples of 10 Avila insurance agents each are obtained. Find the probability that the average number of insurance agents with ADI of the forty samples is between 2.3 and 2.5. [3]

(v) Explain, stating a reason, how increasing the number of samples of 10 Avila insurance agents each will affect your answer in part **(iv)**. [2]

11 **(i)** Sketch a scatter diagram, that might be expected when x and y are related approximately, for each of the cases (A) and (B) below. In each case your diagram should include 6 points, approximately equally spaced with respect to x, and with all x- and y- values positive.

(A) $y = a + bx^2$, where $a, b \in \mathbf{R}^+$

(B) $y = c + \dfrac{d}{x}$, where $c \in \mathbf{R}^+$ and $d \in \mathbf{R}^-$. [2]

Obesity is becoming increasingly prevalent across the globe. To investigate the effects of obesity on one's health, a study was conducted to determine if the blood pressure of adults aged between 40 and 50 years old is dependent on their Body Mass Index (BMI). Data from six patients in this age-group from a hospital are collected. Their BMI, m kg/m^2, and systolic blood pressure, s, in mmHg, are as follows.

m	22	27	31	36	40	44
s	120	150	168	172	179	183

(ii) Sketch the scatter diagram for these values, labelling the axes clearly. [2]

(iii) With reference to your scatter diagram in part **(ii)**, explain why model (B) in part **(i)** is more appropriate than model (A) for modelling these values and calculate the product moment correlation coefficient for model (B). [2]

(iv) Find the equation of the least-squares regression line of s on $\dfrac{1}{m}$ and use it to estimate the BMI of another patient (of a similar age profile) whose systolic blood pressure is 110 mmHg. Comment on the reliability of your estimate. [3]

(v) Explain why the regression line of $\dfrac{1}{m}$ on s should not be used for your calculations in part **(iv)**. [1]

(vi) State, in context, a limitation of using the regression equation in part **(iv)** to estimate the systolic blood pressure of *other* people with BMI within the range $22 < m < 44$. [1]

(vii) Suppose a new data pair $(\bar{m}, \bar{s})$ is added to the table above, where $\bar{m}$ and $\bar{s}$ are the mean BMI (in kg/m^2) and the mean systolic blood pressure (in mmHg) of the adults in the study respectively, based on the data above. Without any calculations, explain whether the equation of the regression line you have obtained in part **(iv)** would change. [1]

– END OF PAPER –